

Effect of a dietary portfolio of cholesterol-lowering foods given at 2 levels of intensity of dietary advice on serum lipids in hyperlipidemia: a r...



CONTEXT: Combining foods with recognized cholesterol-lowering properties (dietary portfolio) has proven highly effective in lowering serum cholesterol under metabolically controlled conditions. OBJECTIVE: To assess the effect of a dietary portfolio administered at 2 levels of intensity on percentage change in low-density lipoprotein cholesterol (LDL-C) among participants following self-selected diets. DESIGN, SETTING, AND PARTICIPANTS: A parallel-design study of 351 participants with hyperlipidemia from 4 participating academic centers across Canada (Quebec City, Toronto, Winnipeg, and Vancouver) randomized between June 25, 2007, and February 19, 2009, to 1 of 3 treatments lasting 6 months. INTERVENTION: Participants received dietary advice for 6 months on either a low-saturated fat therapeutic diet (control) or a dietary portfolio, for which counseling was delivered at different frequencies, that emphasized dietary incorporation of plant sterols, soy protein, viscous fibers, and nuts. Routine dietary portfolio involved 2 clinic visits over 6 months and intensive dietary portfolio involved 7 clinic visits over 6 months. MAIN OUTCOME MEASURES: Percentage change in serum LDL-C. RESULTS: In the modified intention-to-treat analysis of 345 participants, the overall attrition rate was not significantly different between treatments (18% for intensive dietary portfolio, 23% for routine dietary portfolio, and 26% for control; Fisher exact test, $P = .33$). The LDL-C reductions from an overall mean of 171 mg/dL (95% confidence interval [CI], 168-174 mg/dL) were -13.8% (95% CI, -17.2% to -10.3%; $P < .001$) or -26 mg/dL (95% CI, -31 to -21 mg/dL; $P < .001$) for the intensive dietary portfolio; -13.1% (95% CI, -16.7% to -9.5%; $P < .001$) or -24 mg/dL (95% CI, -30 to -19 mg/dL; $P < .001$) for the routine dietary portfolio; and -3.0% (95% CI, -6.1% to 0.1%; $P = .06$) or -8 mg/dL (95% CI, -13 to -3 mg/dL; $P = .002$) for the control diet. Percentage LDL-C reductions for each dietary portfolio were significantly more than the control diet ($P < .001$, respectively). The 2 dietary portfolio interventions did not differ significantly ($P = .66$). Among participants randomized to one of the dietary portfolio interventions, percentage reduction in LDL-C on the dietary portfolio was associated with dietary adherence ($r = -0.34$, $n = 157$, $P < .001$). CONCLUSION: Use of a dietary portfolio compared with the low-saturated fat dietary advice resulted in greater LDL-C lowering during 6 months of follow-up. TRIAL REGISTRATION: clinicaltrials.gov Identifier: NCT00438425.